

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

*I _D.Manjunadha(192324237) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

Q1)

Backtracking Search for Doctor Shift Assignment

Problem

A hospital needs to assign 3 doctors D1, D2, D3 to 3 shifts Morning, Afternoon, Night.

Constraints

1. D1 cannot work Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. All doctors must be assigned exactly one shift.

Backtracking Search

Step 1: Doctor 1

- Try Morning - Valid

Current:

- D1 - Morning

Step 2: Doctor 2

- Try Morning - Rejected Morning already occupied
- Try Afternoon - Valid

Current:

- D1 - Morning
- D2 - Afternoon

Step 3: Doctor 3

- Try Morning - Rejected
- Try Afternoon - Rejected
- Try Night - Valid

Current:

- D1 - Morning
- D2 - Afternoon
- D3 - Night

Start

Doctor 1 Mrng

Doctor 2
Mrng(NO)

Doctor 2 Afternoon (Yes)

Doctor 1
Mrng(NO)

Doctor 3
Afternoon(no)

Doctor 3 Night (yes)

Q2)

Problem

A robot operates in a 5×5 grid and must move from **Start (S)** to **Goal (G)**.

Constraints

1. Movement allowed: **Up, Down, Left, Right**
2. Cost of each move = **1**
3. Robot cannot pass through **obstacles**

Grid Environment

```
S 0 0 X 0
0 X 0 X 0
0 X 0 0 0
0 0 X X 0
0 0 0 G 0
```

Where:

- **S** = Start (1,1)
- **G** = Goal (5,4)
- **X** = Obstacle
- **0** = Free cell

step 1

start the search from the source node and place it in the queue.

- queue = [(1,1)]
- expand (1,1)
- add (1,2) and (2,1)

updated queue:

[(1,2), (2,1)]

step 2

remove the first node from the queue and explore its valid neighbours.

- expand (1,2)
- add (1,3)

updated queue:

[(2,1), (1,3)]

step 3

continue exploring nodes in fifo order.

- expand (2,1)
- add (3,1)

updated queue:

[(1,3), (3,1)]

step 4

ignore blocked cells and already visited nodes.

- expand (1,3)
- add (2,3)

updated queue:

[(3,1), (2,3)]

step 5

bfs continues level by level.

- expand (3,1)
- add (4,1)

updated queue:

[(2,3), (4,1)]

step 6

explore the next node and add only valid neighbours.

- expand (2,3)
- add (3,3)

updated queue:

[(4,1), (3,3)]

step 7

expand the next node and continue searching.

- expand (4,1)
- add (5,1) and (4,2)

updated queue:

[(3,3), (5,1), (4,2)]

step 8

explore all reachable neighbours.

- expand (3,3)
- add (3,4)

updated queue:

[(5,1), (4,2), (3,4)]

step 9

continue until the goal is reached.

- expand (5,1)
- add (5,2)

updated queue:

[(4,2), (3,4), (5,2)]

step 10

if no valid neighbour exists, move to the next node.

- expand (4,2)
- no new node added.

updated queue:

[(3,4), (5,2)]

step 11

continue expanding nodes in queue order.

- expand (3,4)
- add (3,5)

updated queue:

[(5,2), (3,5)]

step 12

explore valid paths only.

- expand (5,2)
- add (5,3)

updated queue:

[(3,5), (5,3)]

step 13

continue expanding until the goal is found.

- expand (3,5)
- add (2,5) and (4,5)

updated queue:

[(5,3), (2,5), (4,5)]

step 14

when the goal node is discovered, it is added to the queue.

- expand (5,3)
- add goal (5,4)

updated queue:

[(2,5), (4,5), (5,4)]

step 15

expand the goal node. the search terminates successfully.

- expand goal (5,4)

S(1,1)

(2,1)

(3,1)

(4,1)

(5,1)

(5,2)

(5,3)

G(5,4)

Path Cost

- Total Moves = 7
- Cost per Move = 1

Total Cost = 7

3Q)

Problem:

The disaster-affected building is represented as a grid where:

- **S** = Start position of the rescue robot
- **G** = Survivor's location (Goal)
- **X** = Blocked path (Obstacle)
- **R** = Risky zone (Extra cost = 2)
- **0** = Safe path (Cost = 1)

Constraints:

- The robot can move Up, Down, Left, Right.
- Some rooms are blocked by obstacles.
- The robot can only observe adjacent cells.
- Each movement costs 1.
- Entering risky zones adds an extra cost of 2.
- The robot must minimize the total travel cost.
- The robot updates its knowledge dynamically while moving.

(a) Analysis of Constraints

Constraint Effect on Decision-Making

Movement only in four directions The robot can explore only adjacent rooms.

Obstacles cannot be crossed The robot must find an alternate path around blocked rooms.

Limited sensing capability The robot has incomplete knowledge and discovers the environment while moving.

Movement cost = 1 Every movement increases the total path cost.

Risky zone additional cost = 2 The robot avoids risky areas whenever possible to reduce total cost.

Dynamic environment The robot continuously updates its map and changes its path when new information becomes available.

(b) Solution Approach

Search Strategy Used

Uniform Cost Search (UCS) with an Online Search Agent

Steps Involved

Step 1: The robot starts from the initial position S.

Step 2: It observes only the neighbouring cells (limited sensing).

Step 3: It assigns the movement cost:

- Normal cell = 1
- Risky cell = 3 (1 movement + 2 risk cost)

Step 4: Uniform Cost Search selects the path with the lowest cumulative cost.

Step 5: Whenever new obstacles or risky zones are detected, the robot updates its internal map.

Step 6: The robot recalculates the lowest-cost path if necessary.

Step 7: This process continues until the robot reaches the survivor G.

(c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an Intelligent Agent because it:

- Perceives the environment using sensors.
- Processes information.
- Makes decisions.
- Performs actions to achieve the goal.

2. Rationality

The robot behaves rationally by:

- Selecting the safest path.
- Minimizing travel cost.
- Avoiding risky areas whenever possible.
- Updating decisions based on newly discovered information.

3. Environment Type

Property	Description
Partially Observable	The robot sees only nearby rooms.
Dynamic	The robot continuously updates its knowledge while moving.
Sequential	Every movement affects future decisions.
Discrete	The building is divided into separate grid cells.
Single Agent	Only one rescue robot performs the task.

(d) Justification of the Chosen Approach

The combination of Uniform Cost Search (UCS) and an Online Search Agent is the most suitable because:

- UCS always finds the minimum-cost path.
- It considers both movement cost and additional risky-zone cost.
- The Online Search Agent can make decisions even without complete knowledge of the environment.
- It dynamically updates the path whenever new obstacles or risky areas are discovered.
- This approach ensures both safe navigation and optimal rescue operations.

Advantages

- Finds the least-cost path.
- Handles risky areas effectively.
- Adapts to unknown environments.
- Suitable for real-time rescue missions.
- Improves navigation safety.